COMPUTATIONAL REPRODUCTION + LAB READINESS

Willsey et al. finger-BCI study | undergraduate research practicum

10 TECHNICAL WEEKS + 1 CAPSTONE

6-8 hours/week | Python/Jupyter | released data, code, and pretrained checkpoints

GOAL

Reproduce selected offline analyses and figures rigorously, then convert genuine technical ownership into an honest application portfolio for the Willsey Lab or a related neural-engineering group.

TECHNICAL MASTERY

- Conda or venv setup, Jupyter kernels, Git, hashes, and provenance.
- MAT inspection, multirate clock alignment, interpolation, segmentation, and indexing.
- Behavioral metrics, nested observations, uncertainty, and unit-of-analysis limits.
- Neural dimensionality, pretrained PyTorch inference, and normalized cross-correlation.
- Directional SNR, channel scaling, train-only PCA, leakage checks, and exact RNG streams.
- Classification, confusion matrices, negative controls, and 3-D flight reconstruction.
- Clean reruns, discrepancy analysis, limitations, and evidence-bounded reporting.

WEEKLY ARC

WEEK	FOCUS
1	Environment, scope, preregistration, provenance
2	MAT schema, clocks, alignment, velocity tests
3	Trial segmentation and traceable Fig. 1c
4	Behavioral functions and Fig. 1e
5	Statistics, nesting, and sensitivity analysis
6	Neural dimensionality and Fig. 2a
7	Released decoder inference and Fig. 2c-d
8	Directional SNR, leakage, and Fig. 3b-c
9	Classification controls and Fig. 4c
10	Clean rerun, discrepancies, technical defense
11	Portfolio, professor questions, mock interview

HOW THE STUDENT WORKS

Each module follows: predict -> specify -> implement -> test -> apply -> compare -> interpret -> disclose assistance -> defend orally. AI and search are allowed, but the private export/search record must name an independent check. The student must explain and modify retained code without AI. There is no point-based rubric.

FINAL TECHNICAL EVIDENCE

- Clean executable notebook, tested functions, environment record, and provenance manifest.
- Selected reconstructions of Figs. 1c, 1e, 2a, 2c-d, 3b-c, ED Fig. 1b-c, and Fig. 4c.
- Full-precision JSON/CSV results, discrepancy log, final report, and live technical defense.

APPLICATION CAPSTONE

- Public-safe repository README and one-page technical brief.
- Five-slide talk, evidence-based CV entry, and tailored inquiry draft.
- Two-minute pitch, evidence-to-claim map, and one-page follow-up proposal.
- Five candidate questions for Dr. Willsey; select one scientific and one research-practice or mentorship question with genuine follow-ups.
- Thirty-minute mock interview covering technical depth, research fit, human-data ethics, skills gaps, and overclaim repair.

PROFESSOR-QUESTION STANDARD

A strong question is grounded in a current source or completed analysis, asks for scientific judgment not already online, fits into about 20 seconds, avoids assuming an opening, and creates a natural follow-up. The student documents why the answer matters before selecting the final two.

COMPLETION STANDARD

The student can regenerate the work, trace central numbers to source data and code, explain assumptions and limitations, repair an unseen error, propose a valid next analysis, and defend every public or application claim. Completion does not imply lab affiliation, endorsement, admission, or an available position.

SOURCE AND SCOPE

Paper: 10.1038/s41591-024-03341-8 | Data: Dryad 10.5061/dryad.1jwstqk4f, version 3

Code: WillseyBCILab/BCI_Finger_Decoding_Quadcopter at commit addd54935c84db611d2e181b3f61d9c60aa9c412

Scope: offline computational reproduction - not acquisition, decoder training, participant interaction, or clinical replication.